MID-TERMEXAMINATION

(CBCS)

B. Tech (CSAI), 2nd Semester

(June, 2022)

Subject Code:: BAI 104	Subject: Introduction To Data Science

Time : 1 ½ Hours **Maximum Marks : 30**

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1			**(2.5*4)**
	(a)	Write syntax to calculate mean of variable **sales** in DataFramedf using NumPy and Pandas. Is there any difference in output of pandas and output of NumPy?	
	(b)	DataFramed having a variable Volume. Is there any differences in output in p1 and output p2 ? (One or two sentences) p1: d=file["Volume"] type(d) p2: d=file[["Volume"]] type(d)	
	(c)	DataFramedf havingn observation and m variable. Consider following one line of code: for index, row in df.iterrows(): what is range of index?	
	(d)	Write program using pandas Consider following list: stock_symbols = ["IBM","ORCL","MSFT", "AAPL"] stock_prices = [116.86, 56.91, 216.51, 119.26] number_shares = [50, 100, 50, 100] Create three series (stock_symbols, stock_prices, number_shares) and then create aDataframedf. Create a new variable amount. The price of a share multiplied by the quantity of shares would get us anamount.	
Q2			**(10)**
	(a)	We haveDataFrame (df) and one of variable sales data as follows: sales 0 1.95 M 1 6.70 M 2 842.16 K 3 1.37 M 4 478.98 K ... Symbol K= 1000 and M= 1000000: sales value 842.16k=1.95*1000 Another variable named company. The majority of companies in theDataFrame have either Group or Corp words in their company names. Write pandas program to find average sales of company having group word in company name?	

Q3			(6+4)
	(a)	We have two datasets (excel file) named df1 and df2 having details of students CSEAI-I and CSEAI-II. We areinterested in finding distribution of variable named marks (marks : total marks secured out of 100) for combined CSAI batch. Which visualization will suitable for this problem and how will approach using tableau? Also you have to color if secured marks is>45 with label 'Pass' else 'Fail' (different color).	
	(b)	We haveDataFrame (df) and one of variablenamed weight data as follows: weight 0 0.93% 1 1.15% 2 0.27% 3 0.98% 4 0.77% ..... Write a panda's program and use lambda function to clean % from data and find sum of weight variable.	
Q4			(10)
	(a)	Wehave created a data frame (by reading a excel file) using pandas named df having many variables (numeric and object). Wehave toreplace all missingvalue in object variable with 'miss' exceptsegmentvariable. Your task is to automate (create a functionand pass dataset nameand which variable name not to replace missing value asparameter, ex: abc (df,segment) so that your program will automatically detect all object variable and replace the missing values. Use for loop foriteration of all object variable.	